

short activity

Modelling the first 20 elements

When the first Periodic Table was devised in the late 1800s, chemists thought that atoms were solid balls that could not be broken down further. No one knew that the atom itself contained particles! So chemists did not know what caused the patterns and similarities in the properties of the elements.

You need

- ☐ Atoms kit or copies of BLM 1.5
- ☐ Wide bench space

What to do

- Step 1:** Divide into small groups of three or four.
- Step 2:** Each group should create a model of the uncharged atoms of each of the first 20 elements. If you have paper models, then these should be cut out.
- Step 3:** Arrange your 'atoms' as they are located in the Periodic Table.
- Step 4:** Examine the electron arrangements of the atoms of the elements that are in Group 1. Identify any similarities. Is there any connection between the electron arrangement and the group number? Does this apply to other groups as well?
- Step 5:** Examine the electron arrangements of the atoms of the elements that are in Period 2. What do you notice? Now examine the atoms in Periods 1 and 3. Can you identify a pattern?

What did you discover?

Copy and complete the following sentences, then add any other conclusion you have drawn.

- 1 The atoms of the elements that are in the same group have the same _____.
- 2 The atoms of the elements that are in the same period have the same _____.

What do you think?

- ☐ From your observations, what might be the prime factor that influences the properties (behaviour) of elements? Justify your answer.
- ☐ How could you identify which sets of elements would be a similar size?
- ☐ Predict what will be true of the radius and weight of the atoms of elements going down a group. Justify your answer.

 Find out on the web

>> Go to www.cengage.com.au/ScienceWA10 and link to **Periodic Table 2**. Pass the hand over the elements to see their names. Click on four of elements from different groups and find out how they got their name, where and when they were discovered, and one other interesting fact about them.

Exercise 02

Answer the following questions in your journal.

- 1 Use your Periodic Table to identify the elements from the following clues. State their names and chemical symbols.
 - a It is a noble gas and its atoms have three occupied electron shells.
 - b It is a halogen that has four occupied electron shells (Figure 1.10).